



Entha Group

Industrial robots, custom machinery and automation — designed and built by one team.

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Entha Group is a team of three mechanical engineers working on the design and fabrication of industrial machines, robots and custom equipment. Our work covers the whole chain: concept and detailed design, engineering analysis, fabrication and assembly, quality control and commissioning.

Our backgrounds complement each other: machine and robot design, mechatronics and automation; manufacturing, composites and quality control; and mechanism design, control and medical robotics. Between us we hold degrees from four of Iran's leading technical universities.

Team



Hassan Nasiri Khonsari

Technical & Design Director

MSc Applied Design, Sharif University of Technology · BSc Mechanical Engineering, IUST

Mechanical design of industrial machines and robots, carried through fabrication, assembly and commissioning. Laureate of the 64th Alborz Elite Prize and twice second-place winner at the Khwarizmi Youth Festival.



Alireza Pirhaji

Manufacturing & Quality Lead

MSc Manufacturing Engineering, Tarbiat Modares University · BSc Mechanical Engineering, Sharif University of Technology

Fabrication of industrial parts and structures, composites and 3D printing. Led production of the postal sorting robots and now runs contract fabrication, quality control and assembly of industrial part sets.



Seyed Mostafa Banihashemi

Project & R&D Lead

MSc Applied Design (Dynamics, Control & Robotics), Sharif University of Technology · BSc Mechanical Engineering, K. N. Toosi University

Mechanism design, control and medical robotics. Handles estimating, procurement and fabrication follow-up for the group's projects. Named top researcher of Sharif's mechanical engineering faculty in 2025.

What we build



Machines & mechanisms

Custom machines from concept design to fabrication, assembly and commissioning.



Robotics & automation

Industrial and service robots, PLC and HMI, electronics and motion control.



Welded structures

Steel and 316 stainless frames for mounting and holding equipment.



Pressure vessels

Enclosures and vessels designed and built for their working pressure.



Injection moulds

Design and machining of plastic injection moulds.



3D printing & composites

Polymer and resin additive manufacturing, composite parts and printer design.



Dental devices

Feasibility and production of aligners, veneers and dental parts.



Models & mock-ups

Display and teaching models, including cutaway assemblies.



Consulting, QC & estimating

Technical consulting, dimensional and weld inspection, techno-economic estimates.

Selected projects

Portraits next to each item show who did the work.

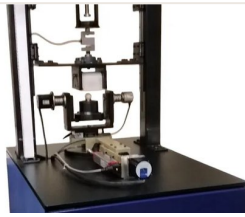


Postal sorting robots



2020–2025

Built and assembled generations 1 and 2 for Iran Post — 50 robots for Tehran and 30 for Kermanshah. Generation 3, designed by Hassan Nasiri Khonsari, went into production at 200 units.



Knee prosthesis constraint & stability test rig



2021

Six-degree-of-freedom rig for testing total knee prostheses to ASTM F1223, the first of its kind in Iran; mechanics, automation, PLC and HMI all developed in-house.



DMD powder feeder



2019

Powder feeder for laser direct metal deposition with complete electronics and control; feed rate controlled from 1 to 100 g/min.



Industrial FDM 3D printers



2020–2023

Five industrial FDM printers designed from scratch, up to a 500 × 500 × 400 mm build volume with ball screws and linear rails on every axis.

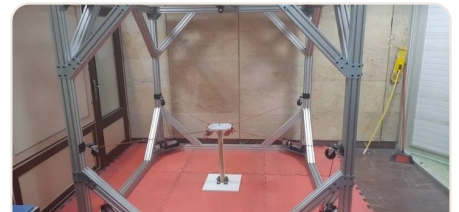


Rail-guided platform over a test pool



2024

Rail-guided mobile platform and cabin over a water-treatment test pool: structure, fabrication and on-site installation.



Cable-driven rehabilitation robot



2022–2025

Cable-driven robot to improve postural alignment in children with cerebral palsy: design, fabrication and control (MSc thesis, Sharif).



Waiter robots



2021

A fleet of seven restaurant waiter robots; chassis and drive designed to run steadily on uneven floors.



Dual-purpose industrial CNC



2023

Modular CNC for wood and light metals with a 100 × 100 cm bed; a head swap turns it into a 3D printer.

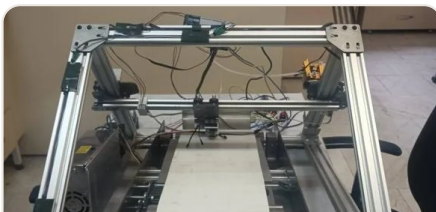


Dental aligners & veneers



2024

Orthodontic aligners and removable dental veneers produced by resin (DLP) 3D printing.



Conveyor-belt FDM 3D printer



FDM printer with a conveyor-belt bed for continuous printing of long parts and batches.



Plastic injection mould



2020

Design and machining of a plastic injection mould.



Jet engine cutaway model



3D-printed cutaway model of a jet engine for display and teaching.



Cable-driven drawing robot



2020

Cable-driven drawing robot that turns any photo into a line drawing on a vertical canvas.

Honors & rankings



Laureate, 64th Alborz Elite Prize



One of 12 laureates in the Technologists & Inventors category, for the knee prosthesis constraint and stability test rig; with a commendation from Sharif University of Technology and a silver medal — 2026.



2nd place, 26th Khwarizmi Youth Festival



For the postal sorting robots — 2024.



2nd place, 24th Khwarizmi Youth Festival



For the knee prosthesis test rig, with a UNESCO certificate and a presidential commendation — 2022.



Best technological thesis, Sharif



Faculty of Mechanical Engineering, Sharif University of Technology — 2021.



Top startup, Sharif Science & Technology Park



For the postal sorting robots team — 2024.



Best design, 4th Iran Industrial Design Festival



For the postal sorting robots — 2024.



Top researcher & technologist, Sharif



MSc level, Faculty of Mechanical Engineering, Research Week — 2025.



Best paper, ICRoM 2025



International Conference on Robotics and Mechatronics, IUST — 2025.



Top teaching assistant



Iranian Society for Engineering Education — 2025.



Rank 16, national MSc entrance exam



Among more than 10,000 candidates; and rank 11 in the final stage of the national student olympiad in mechanical engineering — 2018.



Rank 10, national student olympiad



Mechanical engineering — 2022; also ranked 1st of 117 in his BSc entry year at K. N. Toosi University.



1st-ranked GPA, MSc Manufacturing



Tarbiat Modares University; selected by the exceptional talents office.



Author, “Physics Olympiads in Iran”



Khoshkhan Publishing — 2018.

Publications



Frictional behavior between bone and additively manufactured Ti6Al4V implants is affected by bone density and surface texture under varying loads

Scientific Reports (Nature, Q1), 2025 — DOI 10.1038/s41598-025-25586-0

Our role: development and fabrication of the friction test apparatus.



Large deformation of shape-memory polymer-based lattice metamaterials

International Journal of Mechanical Sciences (Elsevier)



Prediction of Lower Limb Gait Trajectories in Patients with Cerebral Palsy Using Deep Learning Models

ICRoM 2025 — Best Paper



Application of Deep Learning Models in Classification of Sagittal Gait Patterns Based on Rodda's Classification System in Patients with Cerebral Palsy

ICBME 2024



Simulation and Fabrication of Auxetic Polymer Tube and Shape Memory Behavior

9th International Conference on Manufacturing Engineering, Tehran



Design and Control of a Cable-Driven Parallel Robot for Postural Alignment Improvement in Cerebral Palsy

Manuscript in preparation

Professional experience

2025 – present

Design & fabrication contracting — Entha Group · Tehran & Isfahan

Industrial parts, assemblies and structures: estimating and pricing, material procurement, managing machining, laser cutting, bending, rolling and welding, quality control and assembly.

2025 – present

Senior Mechanical Engineer · Tehran

Part and assembly design, dimensional and weld inspection, and management of machining and external fabrication shops; more than 8,000 industrial parts delivered so far.

2025 – present

Production Mechanical Engineer (self-employed) · Isfahan

Contract fabrication, quality control and assembly of industrial part sets; leading an engineering team.

2025

R&D Robotics Engineer · Tehran

Lower-limb exoskeleton: concept design, actuator selection, structure and mechanism, detailed mechanical design.

2024 – 2025

Patent Examiner, Mechanical Engineering — Sharif University of Technology · Tehran

Reviewing and evaluating patent applications in mechanical engineering.

2024

Lecturer — University of Tehran, Kish International Campus · Kish

Taught Manufacturing Processes and Physics 2 in the second semester of 2023–24.

2023 – 2025

Lecturer & Teaching Assistant — Sharif University of Technology · Tehran

Taught the Dynamics of Machinery and Measurement & Control laboratories; TA for Nonlinear Control, Medical Robotics, Musculoskeletal Biomechanics and Advanced Dynamics.

2022 – present

Research Assistant — Medical Robotics Lab, Sharif University of Technology · Tehran

Research in medical and rehabilitation robotics, supervised by Dr Farzam Farahmand and Dr Saeed Behzadipour.

2022 – 2024

Production Supervisor — Postal Sorting Robots · Tehran

Fabrication and production management of smart parcel-sorting robots for Iran Post; production line design to industrial standards; cost and process optimisation.

2022 

Mechanical Engineering Intern — construction 3D printing · Tehran

Concept design, feasibility study and dynamic analysis of a construction-printing robot in ADAMS.

2021 – 2024 

Product Design Manager (part-time) · Tehran

Mechanical and electronic design of an omnidirectional mobile robot.

2020 – 2025 

Head of Mechanical Engineering — Industrial Robotics · Tehran

Mechanical development of industrial robots; managed the build of 50 sorting robots for Iran Post's Tehran distribution centre and 30 for Kermanshah, coordinating workshops and supervising the assembly and integration team.

2019 

Project Intern — Tehran City Theatre automation · Tehran

Automation and stage machinery for the main hall of Tehran City Theatre.

2017 – 2018 

Technical & R&D Specialist — metal forming · Isfahan

Design and assembly of machine parts, thread rolling and cold forging, measurement and quality control, GD&T.

Skills & software

Design & analysis

SolidWorks CATIA Inventor ANSYS Abaqus Mechanism design FEA Vibration analysis

Manufacturing

CNC machining SolidCAM Mastercam PowerMill GD&T Welded structures Cold forging Composites

Automation & control

Delta PLC Siemens TIA Portal HMI LabVIEW MATLAB / Simulink ADAMS PID & sliding-mode control Motion control

Electronics & prototyping

ARM / STM32 AVR / Arduino Altium Proteus FDM & DLP 3D printing

Robotics & AI

Industrial & service robots Cable-driven robots Exoskeletons Deep learning (CNN, LSTM, Transformer)

Contact

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